cBioPortal
FOR CANCER GENOMICS

# Tutorial 4
# Virtual Studies

Create and share your own study from samples of existing studies

Last update: November 12, 2026

cbioportal.org/tutorials

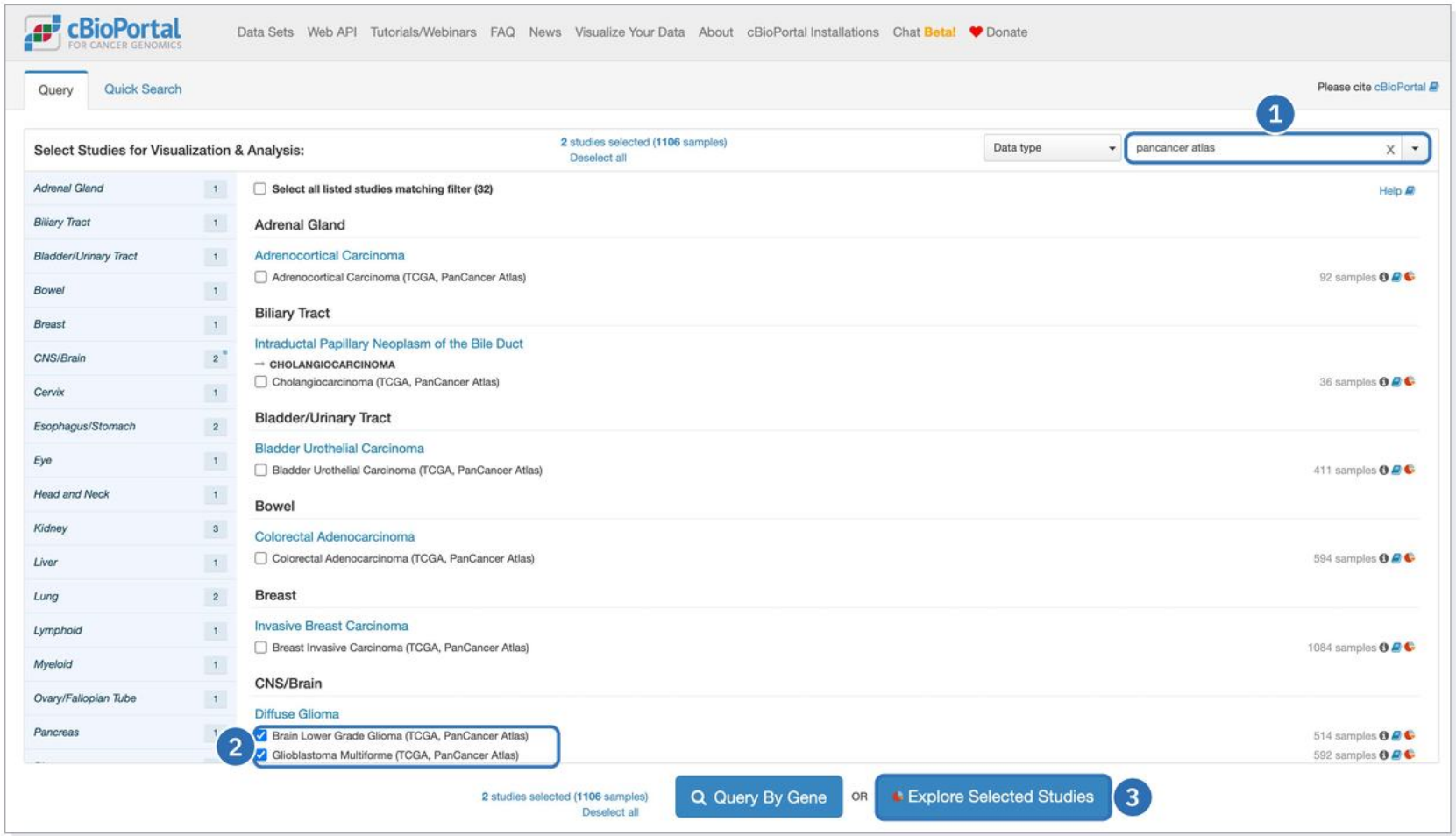
# Tutorial objectives

- Explain what a virtual study is and why it is useful
- Create one: combine studies, select samples, then “Share Virtual Study”
- Choose between a static and a dynamic virtual study
- Show what logging in adds

# What is a virtual study?

- A virtual study is your own study, made of samples from one or more existing studies.
- It can be a subset of one study, or a combination of several, selected by any genomic or clinical criteria.
- You can explore and query it like any other study, come back to it later, or share it with a collaborator.

# Create a virtual study: select the studies

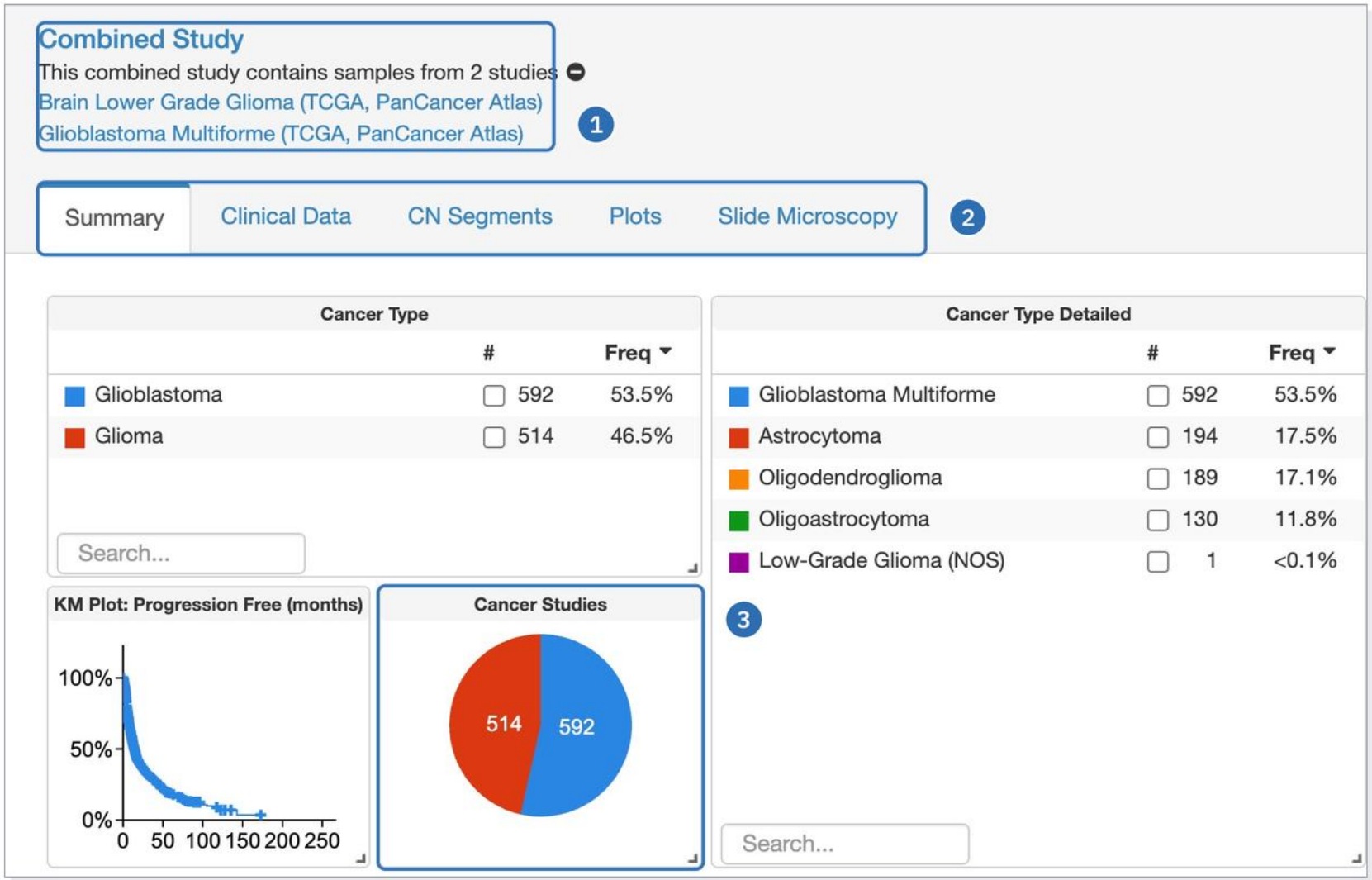



1. Find your studies of interest: type part of their names in the search box. In this example, “PanCancer Atlas”.
2. Tick your studies of interest. In this example, Brain Lower Grade Glioma and Glioblastoma Multiforme (TCGA, PanCancer Atlas).
3. Click “Explore Selected Studies”.



Link to this page

# Create a virtual study: the combined study

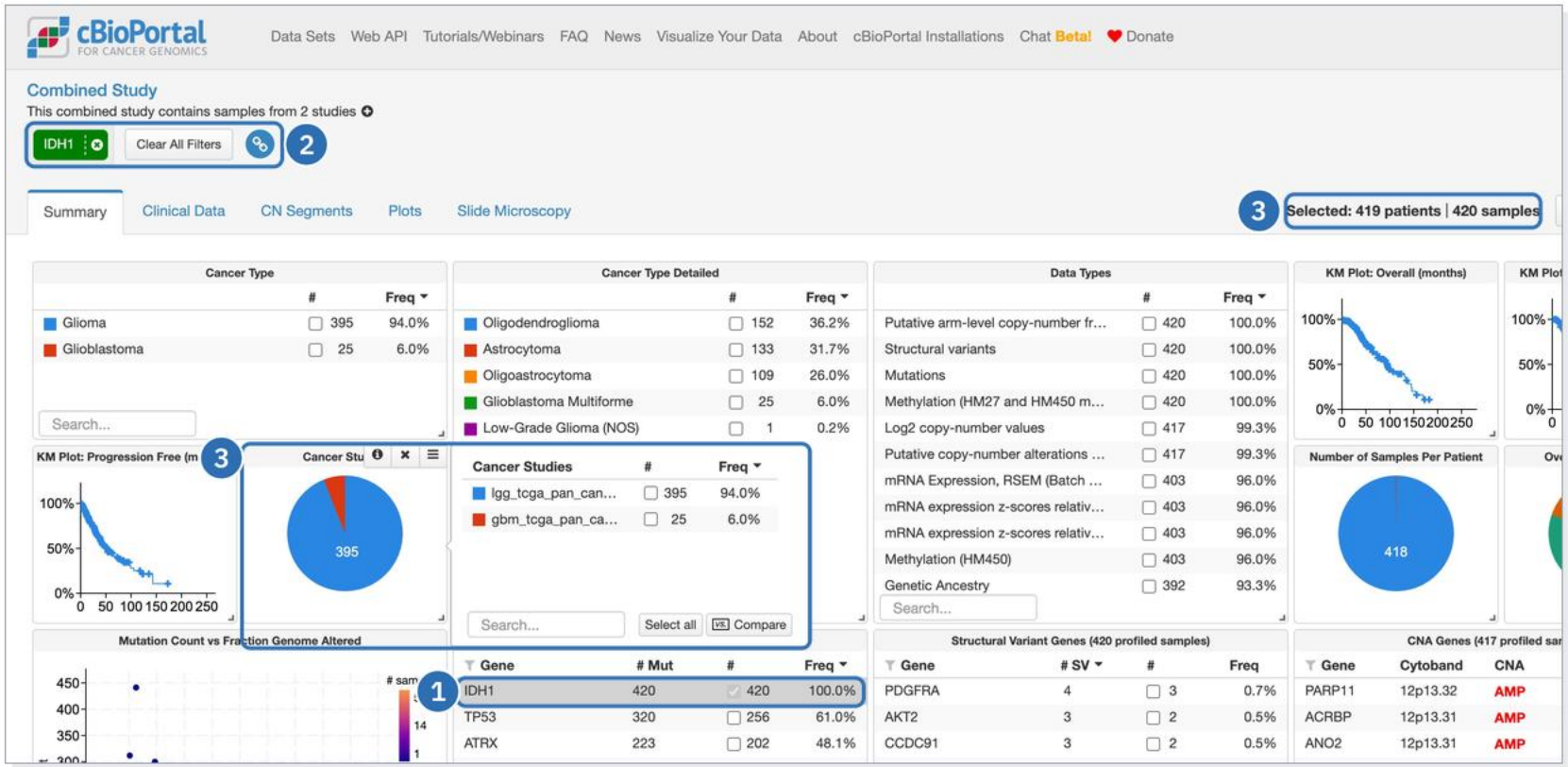



1. A combined study of the 2 studies. Click the + button next to “2 studies” to list them, with links.
2. The same tabs as for a single study.
3. The charts summarize all 1,106 samples. The Cancer Studies chart shows how many come from each study (592 and 514).



Link to this page

# Create a virtual study: select samples

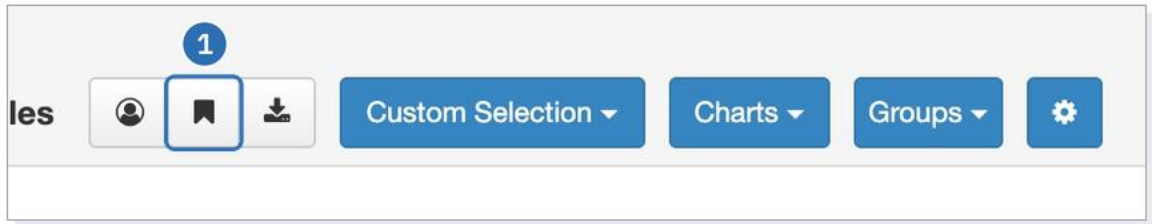


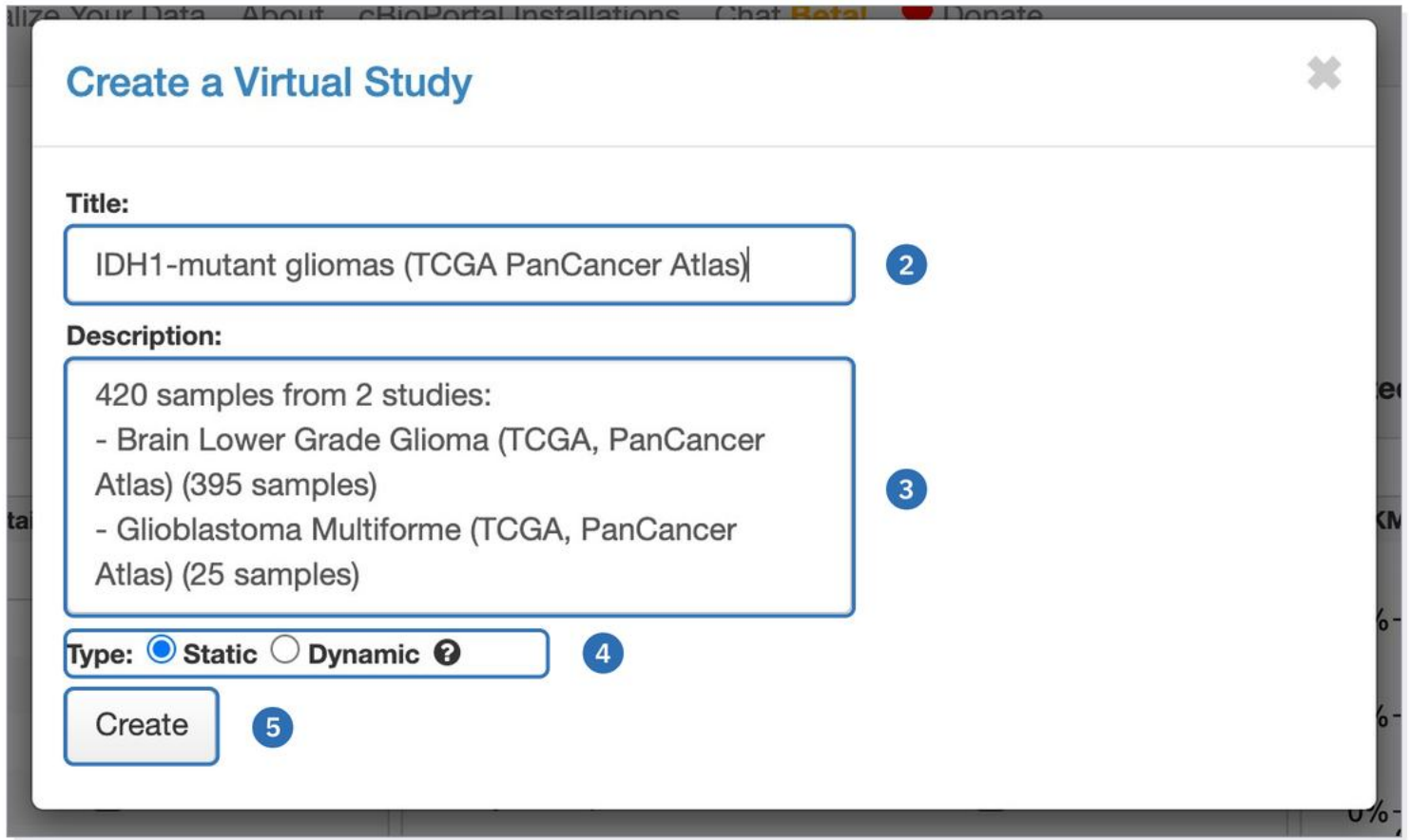

1. Select your samples of interest by filtering on one or more charts. Here we keep only the samples with an IDH1 mutation: tick IDH1 in the Mutated Genes chart, then “Select Samples”.
2. The filter is listed at the top.
3. 420 samples (419 patients) are selected, from both studies: hover over the Cancer Studies chart to see 395 from Brain Lower Grade Glioma and 25 from Glioblastoma Multiforme.



Link to this page

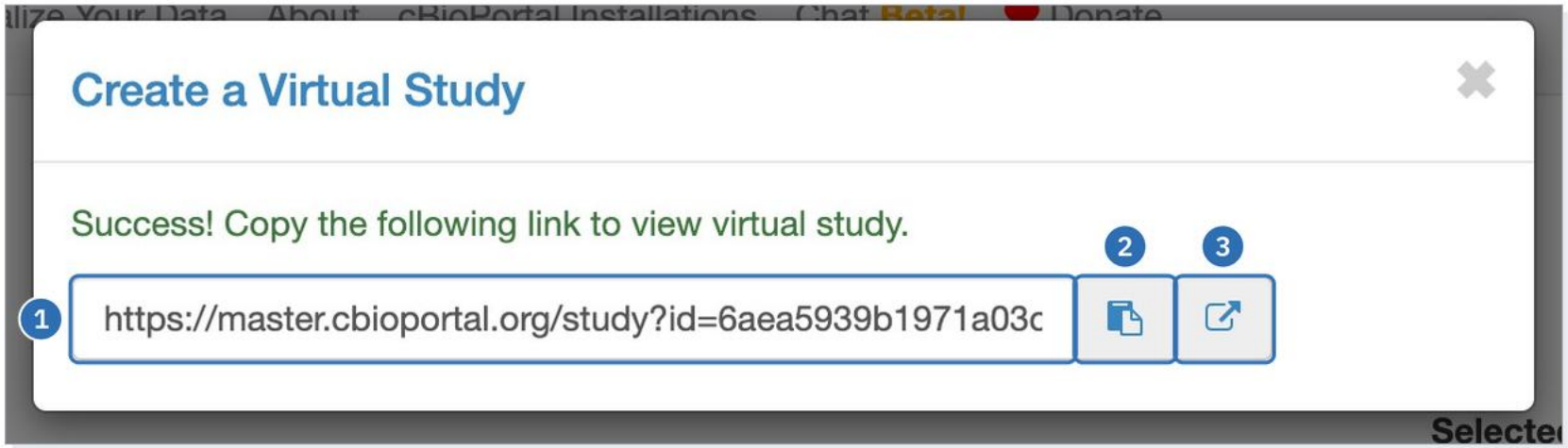
# Share Virtual Study



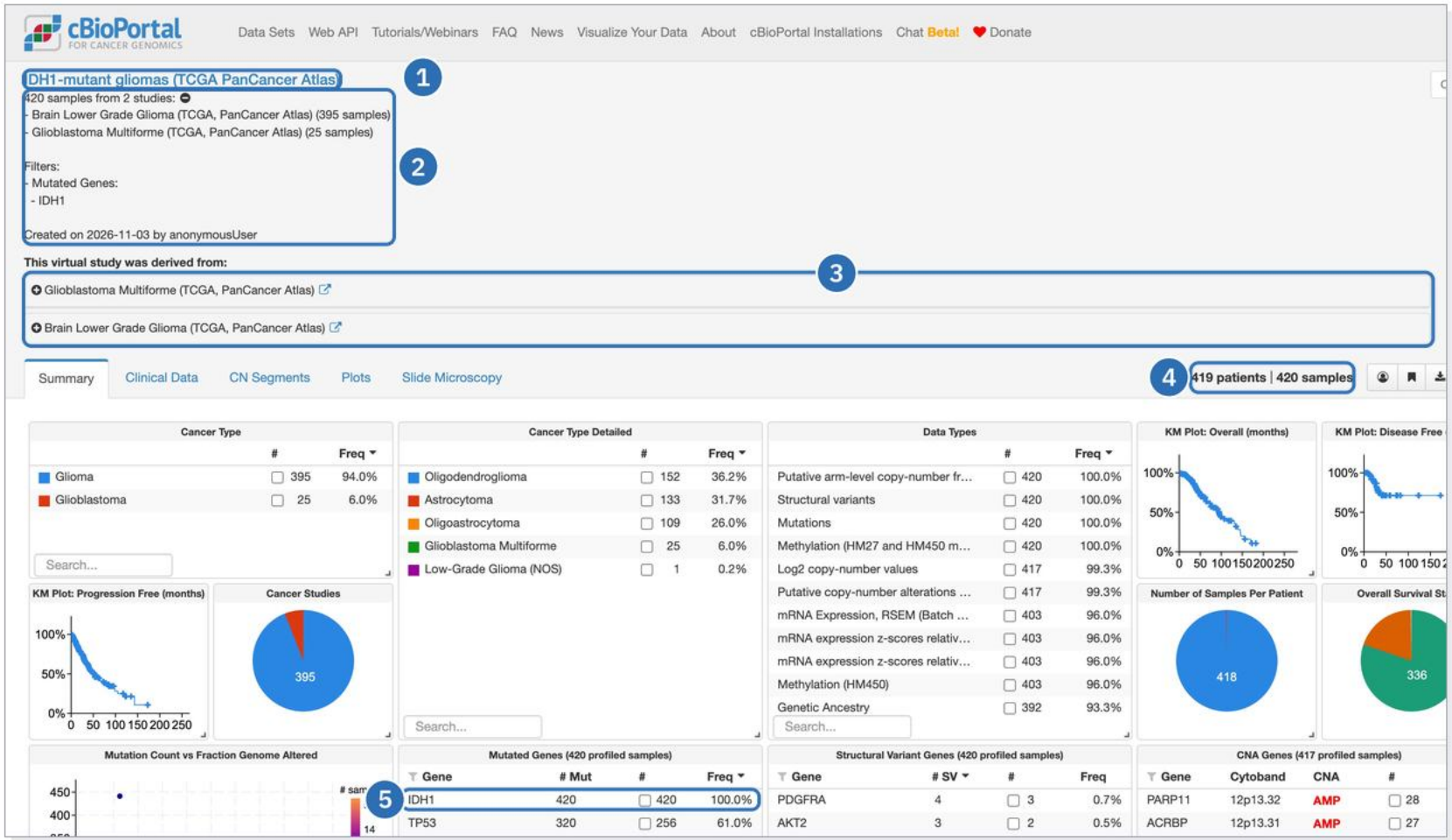

1. Click the “Share Virtual Study” button (the bookmark icon), next to the selection count.
2. Give it a title (optional): here “IDH1-mutant gliomas (TCGA PanCancer Atlas)”.
3. The description lists the studies, the number of samples from each and the filters. You can edit it.
4. “Static”: exactly these samples. “Dynamic”: new samples that match the same filters are added when the studies are updated. The ? button explains both.
5. Click “Create”.

# Share Virtual Study: the link




1. The virtual study is created: this is its link. Links are permanent, so you can bookmark them or send them to anyone.
2. Copy the link.
3. Open the virtual study.

# The virtual study




1. Title.
2. Description, shown with the + button next to it.
3. The studies it was derived from, with links.
4. Only the 420 selected samples are in it: no filter is needed any more.
5. IDH1 is mutated in 100% of the samples. Explore, query and compare it like any other study.



Link to this page

# Do you need to log in?

- No. Anyone can create a virtual study, explore it and query it (with the query box in its Study View), and share its link.
- If you log in, you can also save the virtual study: it then appears at the top of the main page's study list, under My Virtual Studies, and in the query form, and you can combine it with other studies to make new virtual studies.
- Logged in, the “Create a Virtual Study” window also has a “Save” button. Groups (see Tutorial 6) are saved the same way.
- Private cBioPortal installations may log you in through your institution: then there is nothing else to do.




# Questions?

More tutorials: cbioportal.org/tutorials

Documentation: docs.cbioportal.org

Email us: cbioportal@googlegroups.com